

Illustrative groups • Membership does not establish identical samples or independent validation

Family	Representative studies	Interpretation
Design2Code lineage	P01, P05, P06, P07, P24, P67	Benchmark or source-design ancestry; relationships differ across studies.
WebCode2M lineage	P03, P09, P25	Dataset reuse does not multiply independent criterion validation.
Optimization roles	P19, P69	Conceptual grouping; task resources must be checked study by study.
Boundary comparators	P20 FrontCoder, P21 ReLook	Critic/checklist roles; outside confirmed external-visual-target core evidence.
Compatibility extension	P23 with D2C/DCGen ancestry	Related source designs can support a distinct deployment criterion.

Complete typed relationships and within-study dependencies are recorded in Supplement S5 and dependencies.json.